

MSc in Computer Science - Team Project

Interim Report

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1. User Scenario: The Characters

1.1. Background

The family is the basic unit of society (United Nations, 1948). Providing care, support, and stability to our loved ones is not merely out of biological obligation, but because we are bound by mutual care and commitment. Sometimes though, our own resources fall short, and we must rely on outside experts. When those experts fall short of expectations, it adds another layer of stress.

Ireland is facing a projection of growing older demographic. Under the CSO's medium migration scenario, the population aged 65 and over rises from 781,400 in 2022 to 1,399,000 in 2042, an increase of 79%, and the population aged 80 and over more than doubles, from 181,200 to 429,100 (CSO, 2024).

The Economic & Social Research Institute (ESRI) projects that long-stay capacity has to grow from 29,579 beds in 2022 to between 47,588 and 53,266 by 2040 (Walsh and Kakoulidou, 2025). Where beds are scarce, families take the home that has a place rather than the one they would choose by preference, and this falls hardest on families caring for older adults with functional dependencies or cognitive frailties such as dementia. In the nursing home space, at the end of 2025 Health Information & Quality Authority (HIQA) has registered 545 nursing homes with a total of 32,408 beds (HIQA, 2026), and more than 24,000 people were funded under the Fair Deal scheme in 2024 (HSE, 2025).

In the current digital age, information is everywhere. Yet families and older adults still struggle when it comes to arranging long-term care, for a loved one or for themselves. Guides, contacts, and forms are scattered across the web. The Health Service Executive (HSE) and Citizens Information hold most of the information needed, but families have to find their way through these sites on their own.

1.2. Real-life Experience

The team interviewed a potential end user who had recently had difficulty arranging long-term care for a loved one. The interview made the problem clearer: people are not prepared for the situation and are unsure what support is available. The problem is worse after a fall or a medical emergency, when the hospital seeks

an immediate transfer to a nursing home. In practice, families often do not choose a nursing home.

People should be able to choose, and even when time is short, they should be informed. The project aims to help families and older adults understand the process and what to know before choosing a home. How it is presented matters just as much, because families and older adults need it to be clear, easy to follow and usable at a stressful time.

1.3. User Examples

1.3.1. A Family Member's Perspective

Declan is 61 and lives in Limerick. He is arranging care for his mother, who needs high dependency care after a stroke and must move into a nursing home within weeks. He is comfortable with technology but has little time, and he wants to bring a shortlist of options quickly and confidently. He already knows the type of care needed and the area he prefers. His priority is homes that meet her needs and have a strong compliance record. He can find individual HIQA reports, but he cannot filter or compare them, and they are too long to read for several homes in the time he has. He wants to know which homes in his area offer high dependency care, their recent inspection results, and whether any compliance issues reoccur.

1.3.2. An Older Person's Perspective

Michael is 68 and lives alone in Dublin. He has always been independent, but over the past year he has realised that he may need more support with everyday activities. His family has offered to help him look at nursing homes, but he wants to make the decision himself. He wants to know which homes are in areas he would be comfortable living in and what life and care are really like there, before he is forced to decide quickly. He discovers that HIQA publishes inspection reports, but they are long and full of terms he does not immediately understand. He particularly wants to know whether a home has had problems with residents' rights, staffing, care planning or safety, and whether those problems came up again at later inspections. He wants to see the evidence and judge for himself rather than be told which home is best.

1.3.3. User Flow

An initial flowchart (Figure 1) gives users a recommended route through the website, from reading the guides to searching for and contacting a home. The prototype presents a lot of information at once, and later development could turn it into an interactive tutorial that guides users through the site one step at a time.

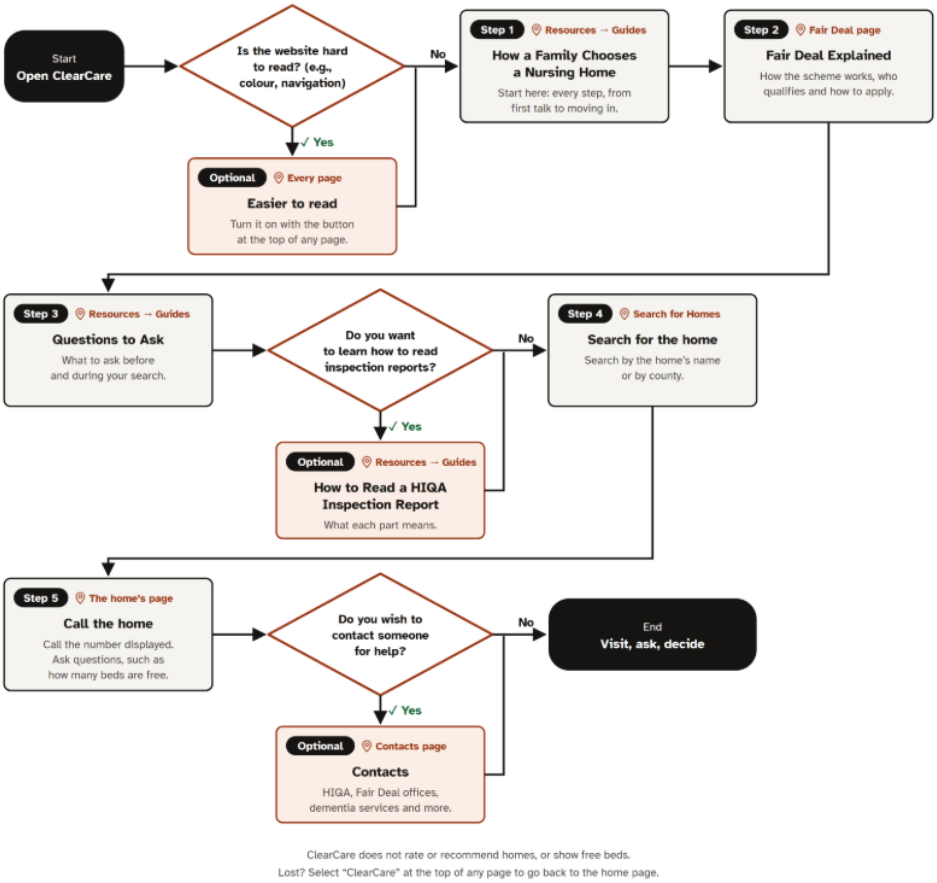


Figure 1. User flow through the site

2. Technical Problem

2.1. Why the System Exists

The information families need exists, but it is spread over several sites and formats. The HIQA website publishes the register and the inspection reports

where each contains a PDF of 20 to 40 pages written for the provider and the regulator. The HSE website publishes the Fair Deal rules as web pages and the weekly prices as two PDF lists. Finally, the Citizens Information website explains

the scheme and does not list homes. A family deciding under pressure must open each report, find Appendix 1, and compare judgments by hand across several years, for every home on their shortlist.

The system does that reading once, for every registered home, and shows the result in plain language: what each inspection found, whether the home fixed what it promised, and how much it costs. Every quotation on the site is the regulator's or the provider's own words, with its date and page, and every derived figure states the reports it was counted from. It does not rate or recommend homes.

2.2. The Core Technical Problem

2.2.1. Back-End

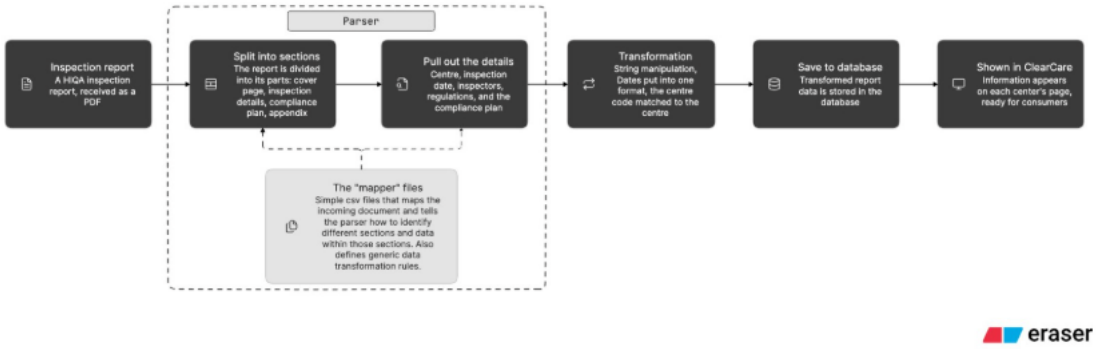


Figure: End-to-End parsing flow diagram

The evidence this project depends on exists only as PDF documents. A PDF stores what was printed and where, not what each part of it means, so nothing in the file says that a block of text is a table, that a line is a heading, or that a row gives a regulation and its judgment. All of that has to be worked out from the layout of the page. The reports are written and exported in different ways, so headings change between templates, tables carry on over a page break with only some cells filled in, and some tables are printed with no lines between the rows at all, which is why the public price list arrives as one row per county instead of one per centre. A parser that assumed a single fixed shape would fail on much of the corpus. What makes the task more than mechanical is that poor results are easy to miss: a document read badly produces empty records, and empty records look the same as a report that found nothing. The parser has to be good enough to handle several

PDF layouts and structures, and to make an incomplete result visible rather than silently empty.

2.2.2. User Interface/ User Experience

Several organisations already publish material on nursing homes, including Sage Advocacy, Nursing Homes Ireland and HIQA, but none of it is written with older readers in mind, and setting the same information out in fewer words is where the design difficulty lies. The project's design work therefore depends on what can be learned from those sites and from the first round of user evaluation. Evaluating the easier-to-read mode will be difficult, since asking older people to examine a nursing home website is an uncomfortable request for many of them, which limits both who can take part and what can reasonably be asked.

2.3. Existing Systems

Findcare.ie was a private directory. When the team reviewed it in September 2026, it listed registered homes with the per-regulation judgments of their latest HIQA report, the same grid that ClearCare's centre page shows. It did not follow a home across inspections or show the provider's commitments. But lately, the findcare.ie service seems to be down and returns "This service has been suspended" message when visiting the website.

The HSE Fair Deal pages explain the scheme: how to apply, the assessments, the three-year cap, the nursing home loan and appeals, with worked examples in euro per week. On choosing a home they offer about a dozen lines of advice and a link to HIQA. The prices of each registered home are stored in two PDF lists with different layouts: 411 private and voluntary homes over 13 pages, ordered within each county by HIQA number, and 115 public homes over 3 pages.

HIQA's website has a search functionality where users can filter by centre type, county and keyword, and each result shows the summary details of the nursing home centre. This includes an address, phone, maximum occupancy, registration details and the last three years of inspection reports as PDF downloads, which appear four to five months after the visit. However, the website does not show a summarization of what the inspectors found, the judgement for each regulation, the provider's compliance plan on the website. The user must download the

report and analyze them themselves. In the case scenario of having to compare two homes, users would have to download both reports and thoroughly read through each which is not ideal.

Each site covers its own part: the inspection evidence on HIQA and the Fair Deal application on the HSE site. No existing system puts them together that allows families to make an informed decision based on one's home inspection to the next.

3. Technical Solution

3.1. What the System Does

ClearCare is a publicly accessible website that brings together the HIQA register and inspection reports for nursing homes in Ireland. It presents this material in a single, searchable form so that residents, families, and other interested readers can find and interpret it more easily.

Search - The platform covers all 545 registered homes. Users can search by name, address, or county. Each result shows the register's recorded address, Eircode, registered capacity, and telephone number.

Centre Pages - Each home has its own page covering inspections that was carried out. For every regulation assessed, the judgment awarded at each inspection is displayed side by side, and changes between inspections are marked as "worse", "same", or "not assessed". Each inspection also carries a count of judgments rated compliant, substantially compliant, and not compliant, with a link to the original HIQA report in PDF form. The page further states the home's type and its maximum contract price as published in the HSE lists.

National Figures - This section looks at what happened to each judgment at the home's next report, drawing on all 1,969 reports currently loaded. The findings are presented as a matrix of next result by starting judgment, alongside the re-check rate for each starting judgment and the coverage on which these figures depends on.

Guides and Contacts - The site includes practical guidance on choosing a home, the Fair Deal scheme, reading an inspection report, and the steps from first

identifying a need to moving in. Contact details are provided for the HSE, HIQA, and the Fair Deal office.

Accessibility and Display - An “easier to read” mode, designed with older users in mind, offers larger type, higher contrast, and larger interactive targets. It can be switched on with a single tap and is remembered across visits. The site also supports light and dark display modes.

3.2. How it Works

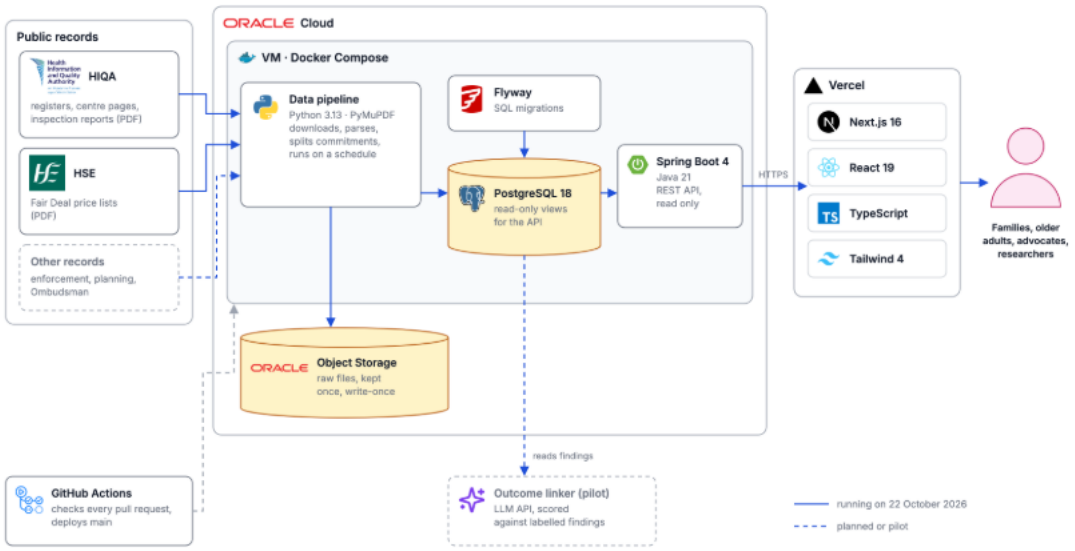


Figure 3. System architecture

3.3. Front End

3.3.1. Technologies and User Interface Components

The system's front-end is built with Next.js and React 19, written in TypeScript. The pages are rendered on the server, which requests data from the Spring Boot API each time a page is loaded, so browser never calls the API directly.

Client-side React is used only for the few components that need to respond in the browser, such as the theme and Easer to read toggles, the hamburger menu for phones, the dialogue that warns users before being redirected to an external site, the carousels and the inspection history scroller.

With TypeScript types, it allows the website to mirror its OpenAPI specifications, so the compiler can catch a misuse of data before the site is deployed.

The design is styled using Tailwind CSS 4, with colours, fonts and sizes as CSS variables in one stylesheet. This allows the theme and Easier to read modes to change those variables rather than restyling each page.

The standard design follows WCAG 2.2 level AA (W3C, 2023). The Easier to read design (Figure 4) is for older adults who find the standard design harder to see (W3C, n.d.) and applies the level AAA criteria for enhanced contrast, visual presentation, focus appearance and emphasized font size and styles.

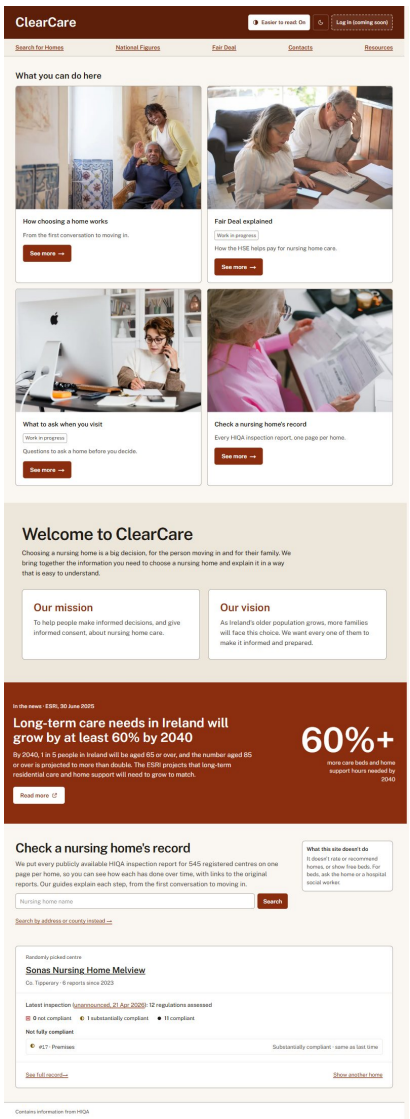


Figure 4. The home page with Easier to read switched on.

3.3.2. Wireframes

The first wireframes were drawn in the early stages of the project with Claude Design, which allowed the team to turn its initial requirements into screens quickly. Wireframe 1 (Figure 5) shows search and shortlisting: users search by name, county or Eircode, each result card shows the latest inspection and publication dates and how many regulations were judged compliant, substantially compliant and not compliant, and a result can be added to a shortlist for the comparison grid of Wireframe 2 (Figure 6). In the grid each row is a regulation and each column a shortlisted home, and each cell is a timeline of that home's inspections: a filled circle for compliant, half-filled for substantially compliant, a cross for not compliant and a dashed circle for not assessed, so that the symbols work for people with colour vision deficiencies and in print. Where a home's inspections cannot fairly be compared with the others, for example one 18 months apart from the rest, the grid says so.

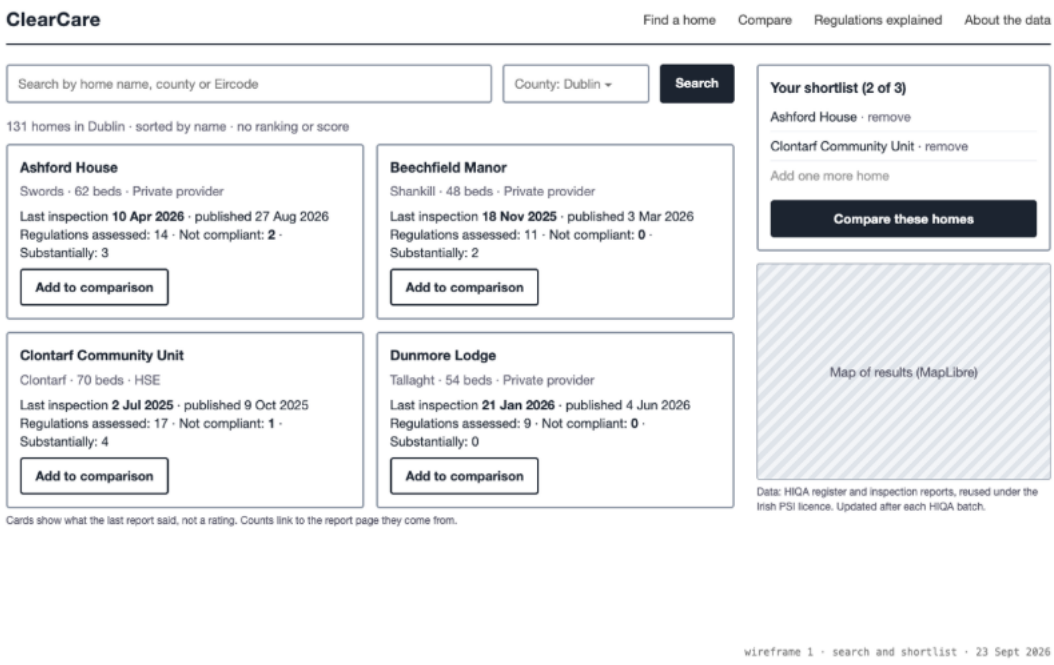


Figure 5. Wireframe 1: search results and shortlist.

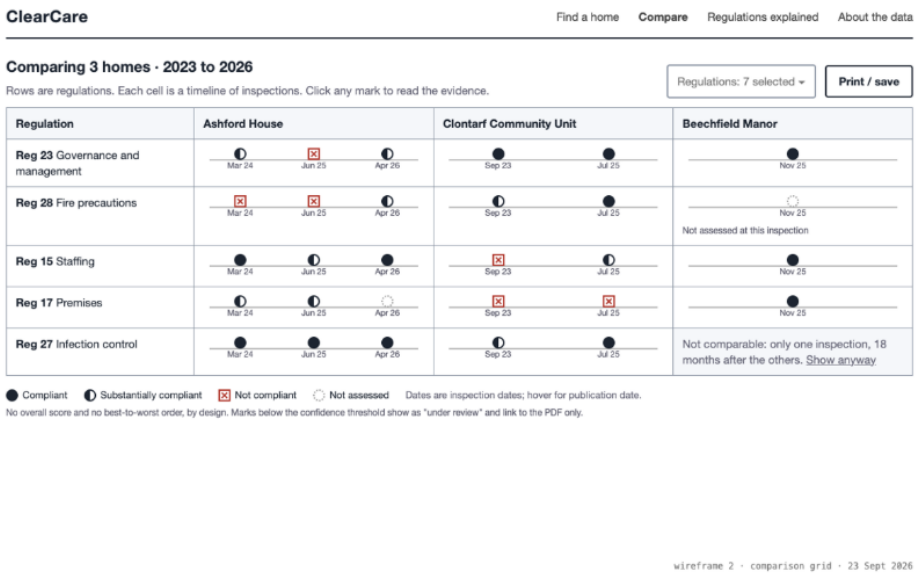


Figure 6. Wireframe 2: comparison grid, one regulation per row and one shortlisted home per column.

The wireframes set a principle that has stayed with the project: the site reports what the inspections say and does not rate homes. The design has been revised in each development's progress since. Figure 7 shows an early version of the home page.

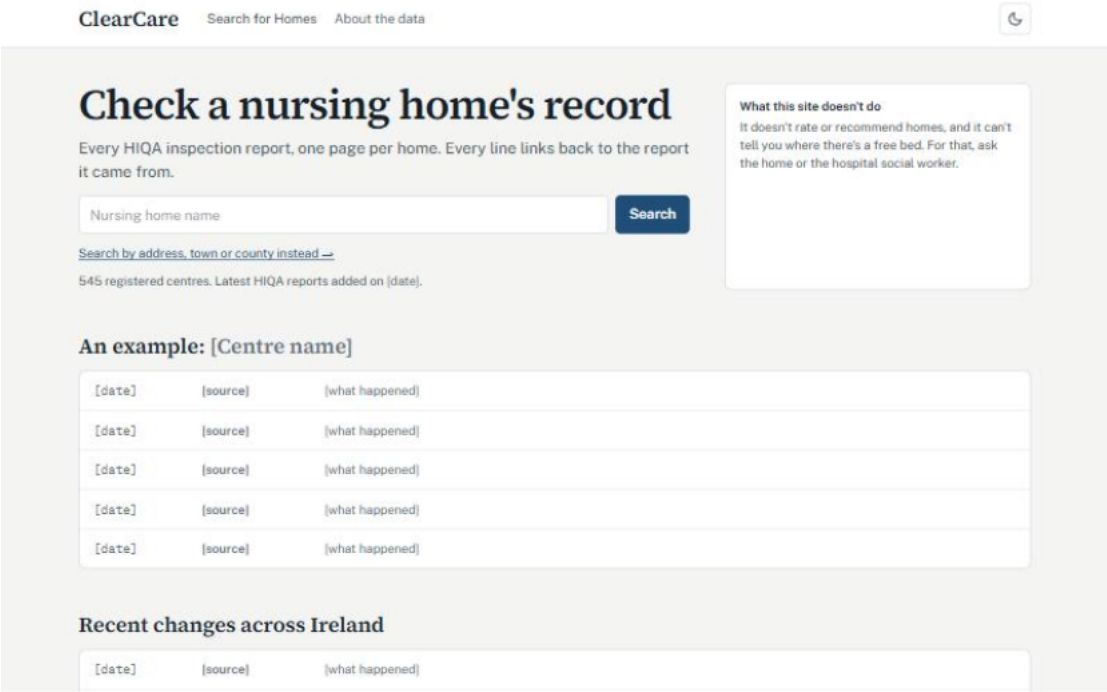


Figure 7. An early version of the home page.

3.3.3. Design Changes

The design has changed considerably over the weeks. The navigation now leads to Search for Homes, National Figures, Fair Deal, Contacts and Resources. As the planned entry point, the home page (Figure 8) is revised to introduce the user with a carousel of guides and a welcome section setting out ClearCare's mission and vision. Drawing from sageadvocacy.ie and nhi.ie, banners and section layouts were used so that users who know those sites would find it familiar.

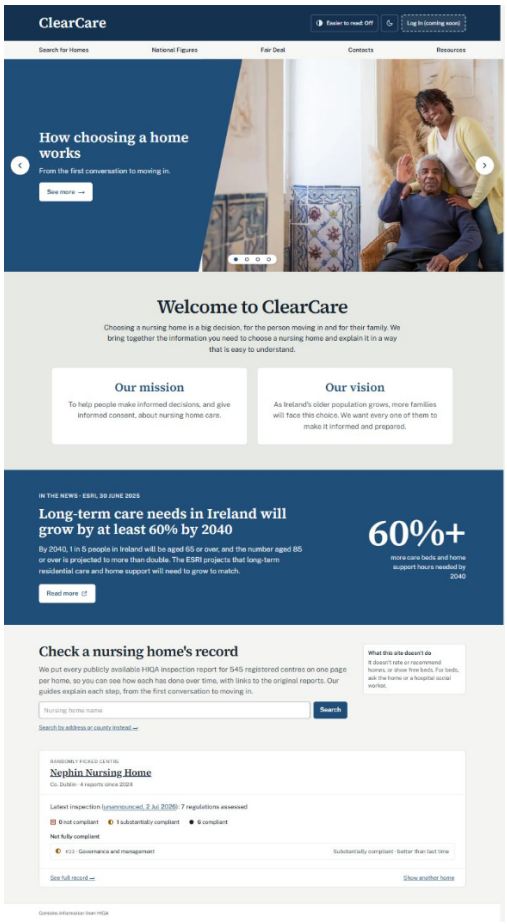


Figure 8. The home page now.

The centre page (Figure 9) shows each home's details, including the full weekly cost, and the summary of its inspection reports with a legend for the judgment on each regulation.

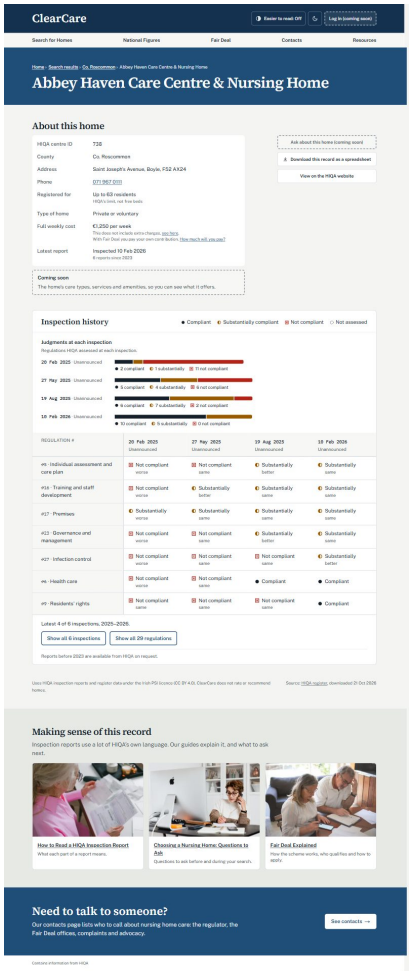


Figure 9. The centre page.

The resources page (Figure 10) brings together the guides: how a family chooses a nursing home, questions to ask, how to read an inspection report, what Fair Deal is, and how to use the website. Some guides are labelled “Work in progress” and will be finalized in later development.

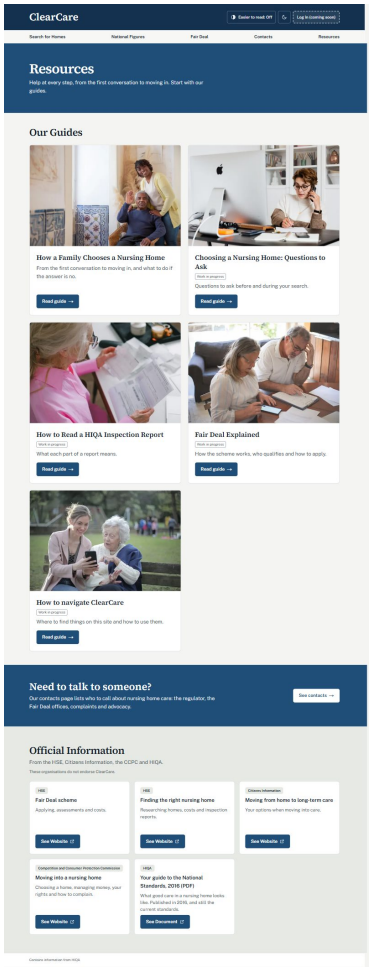


Figure 10. The resources page.

3.4. Back End: Technical Components

Public Records (HIQA, HSE, and other records) - These are the upstream data sources. HIQA supplies registers, centre pages, and inspection reports (PDF), and the HSE supplies Fair Deal price lists (PDF). "Other records" such as enforcement, planning, and Ombudsman data are shown with a dashed border and line, so they are planned rather than live.

Data pipeline - A Python 3.13 service that uses PyMuPDF to download the source documents, parse them, and split them into individual commitments. It runs on a schedule, writes the raw files to Object Storage, and loads the structured results into PostgreSQL.

Object Storage (Oracle) - This holds the raw downloaded files, kept once and write-once. That preserves the original source documents as an unaltered record, so parsed data can be traced back to the original or re-processed later.

Flyway - Manages the database schema through versioned SQL migrations applied to PostgreSQL. This keeps schema changes repeatable and tracked in version control across environments.

PostgreSQL 18 - The central database, which stores the parsed data from the pipeline. It exposes read-only views for the API, so the API only sees a stable, curated layer rather than the underlying tables.

Spring Boot 4 (Java 21) - A read-only REST API that serves data from the PostgreSQL views. Because it can't write, the public-facing layer is simpler and lower-risk. It talks to the frontend over HTTPS.

Outcome linker (pilot) - A dashed, pilot-stage component that reads findings from PostgreSQL and uses an LLM API to link them to outcomes. Its output is scored against labelled findings, which gives a way to measure accuracy before it's relied on.

GitHub Actions - The CI/CD layer, which checks every pull request and deploys main. A dashed line connects it to the Oracle Cloud environment, which suggests the automated deployment path is still planned.

Oracle Cloud and VM (Docker Compose) - The hosting environment for the backend. The pipeline, Flyway, PostgreSQL, and the Spring Boot API run together on a single VM orchestrated with Docker Compose, with Object Storage alongside it. This is a simple, cost-effective setup, with Vercel separately handling the frontend.

3.5. Data

The project draws on two data sources, both published and reached through their public websites.

The first is the set of inspection reports published by HIQA, available at https://www.hiqa.ie/reports-and-publications/inspection-reports. Each report records what inspectors found in a nursing home and how the provider responded to those findings. A scheduled job downloads them on a recurring basis, so the collection stays current without anyone fetching files manually. The downloaded files are stored in object storage on Oracle.

The second source is the maximum contract prices for nursing homes, which the HSE publishes as two separate lists: one for private nursing homes (https://www2.hse.ie/documents/3223/HSE_Section40_List_230_Published_List_04.09.2026.pdf) and one for public nursing homes (https://www2.hse.ie/documents/7322/NHSS_Cost_of_Care_September_2026_Update.pdf). They are published as standalone documents rather than through a feed, so each list is downloaded and processed manually at present. For that reason, they are not yet held in the object storage that holds the inspection reports and bringing them into the same arrangement is the next step for this part of the work.

4. Evaluation: The Reviews

4.1. What Success Looks Like

The system is successful when a person choosing a nursing home, or the relative of a resident, can find a home and judge its record without prior knowledge of inspection terminology: what inspectors recorded, whether the issues were resolved at the next inspection, and the weekly cost. The Easier to read design should let an older reader reach the same understanding in fewer words. The source and date behind every answer must stay visible, so that a reader can verify it.

From a technical perspective, success means that the system sustains these answers without manual work: a new report or price list is incorporated without anyone handling it, each figure can be traced to a stored document, repeating a load writes nothing new, and a corrected publication is kept as a new version beside its predecessor. Failure must be visible: a price list matching no centre, or a report yielding no regulations, is reported rather than shown as an empty page. This project could also act as a pre cursor for a potential dataset collection and curation for anything related to elderly care/HIQA reports. Turning unstructured and scattered reports from HSE, HIQA to structured datasets that can be used for ML modelling, trend analysis, etc.

4.2. User Evaluation

4.2.1. General User Evaluation

In later development, adults will be asked to review the site and list three things they liked, three they did not like, and three they wish had been included.

At the last phase of the project, the team plans to announce the website publicly with a visitor survey on a five-point Likert scale, from 1 for strongly disagree to 5 for strongly agree, with each question derived from Nielsen's usability heuristics (Nielsen, 1994). Because Likert responses are ordinal, each question will be summarised by its median, mode and the percentage who agreed or disagreed, rather than a mean, and the questions grouped by heuristic in a diverging stacked bar chart, so that the lowest-scoring heuristics can be prioritised in later sprints. Written responses and survey comments will be analysed by thematic analysis (Braun and Clarke, 2006): coded and grouped into recurring themes, to identify common strengths, problems and requested features.

4.2.2. End User Evaluation

The team also aims to interview adults who have been through the experience of choosing a nursing home: whether the website was usable and helpful, whether any information was or seemed inaccurate, whether it told them anything new, whether the information was sufficient and easy to understand, whether the site would have helped them before their experience or would help next time, and whether they would recommend it. The interviews will be transcribed and analysed by thematic analysis (Braun and Clarke, 2006), and the themes mapped to four questions: is the website usable, will it stay useful as new reports are published, is it accessible and is it accurate.

4.3. Data Evaluation

The amenities extraction is evaluated against human labels, using a small gold set of reports read and labelled by hand. The model's output is then compared with these labels field by field.

For amenities, a gold set is being built from 18 Dublin centres, using the most recent report for each. The centres were chosen by hand for variety rather than taken in order: they include religious-run and private homes, a multi-unit centre, centres with long inspection histories, a centre in north county Dublin, and centres

whose names ("Hospital", "Care Centre", "Clinic") needed checking against the nursing home category. Any centre that turns out to be a different type is swapped and the swap recorded. A labeller reads each PDF and records the nine amenity fields (care types, room types, outdoor space, activities, transport, religious services, special units, dining notes and other notes) in a spreadsheet, one row per centre, before seeing the output of the language model (GPT-4o-mini). Only what the report states is recorded, so the three yes/no fields are labelled as mentioned or not mentioned rather than yes or no. Since this section of the report is written by the provider, the gold set measures how faithfully the model extracts the report, not whether the home actually has those amenities. Each field is then compared with the model's output and classed as a match, a false negative (in the report, missed by the model) or a false positive (given by the model, not in the report), giving an overall accuracy rate and a per-field breakdown. Scanned PDFs without selectable text are excluded rather than counted as model errors. The gold set is still being built, so no figure is available yet.

5. Research Questions

5.1. Ontology Research: Characteristics that may affect the Maximum Contract Price

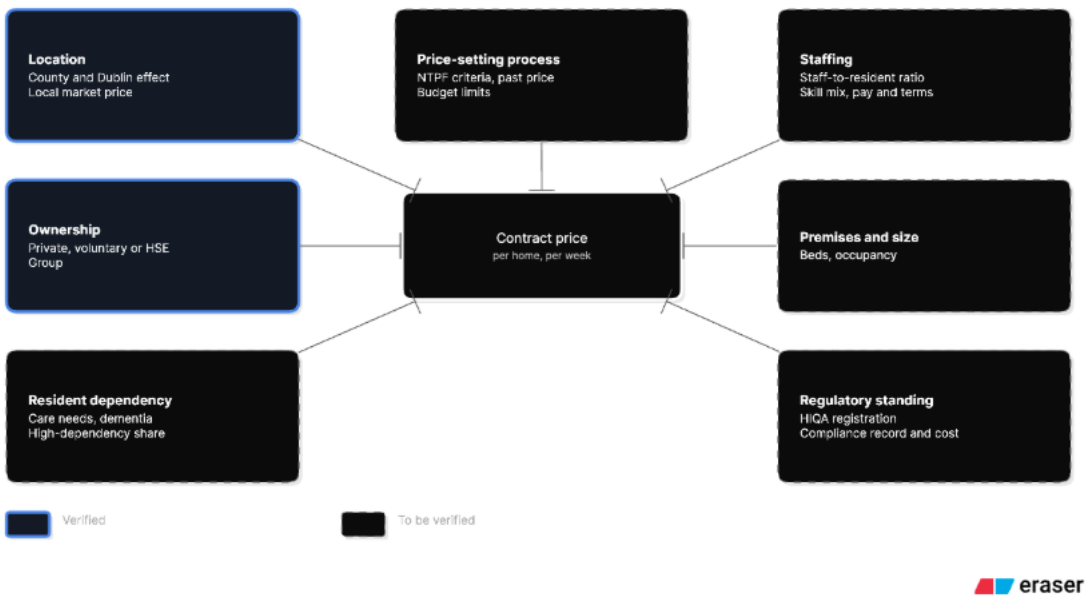


Figure 11. Characteristics that may impact prices

The initial research identifies location as the first factor to examine when analyzing contract prices. Other candidate characteristics are the type of home (private, voluntary or public), registered capacity, and the facilities and services available. The HSE publishes two lists: the maximum agreed prices for 411 private and voluntary homes and the cost of care in 115 public homes, every record with a numeric weekly value. Private and voluntary prices range from €1,192 to €1,508 (median €1,287). Excluding one public home printed at €342, public values range from €1,193 to €2,902 (median €1,966). Dublin homes average about €1,408 a week, compared with about €1,276 in counties outside Dublin.

A review of 16 HIQA inspection reports from 2023 to 2026 also identified recurring areas of inspection: premises, infection control, individual assessment and care planning, and food and nutrition. These are descriptive characteristics of homes. The review does not show that any of them affects the price.

5.2. Predictive Modelling: The Likelihood that a Home Fails a Regulation

The initial audit of 16 HIQA reports identified regulatory judgments, including Compliant, Substantially Compliant and Not Compliant, as well as recurring areas requiring attention. The audit also highlighted the importance of retaining the original inspection wording and source-page references so that each finding can be traced back to the relevant report.

These judgments could be the basis for analyzing compliance patterns across homes and regulations: which regulations are most often found non-compliant, and whether an earlier judgment is associated with a later outcome. The audit alone does not establish a reliable probability that a particular home will fail a regulation. Before that likelihood can be estimated, the extracted judgments must be validated, differences in inspection coverage considered, and the analysis tested against appropriate data.

6. Conclusion: The Plan

6.1. Project Management Strategy

The project follows Scrum over the module's twelve weeks, with JIRA for issue tracking and sprint management. Each sprint is one week long, with a sprint goal

set at planning alongside individual deliverables. A retrospective is held every Friday, followed by sprint planning. The team leader role rotates each sprint and every member reports progress at a daily stand-up. GitHub holds the repository, with a ruleset that forbids direct commits to main and requires at least one approval on every pull request. Communication and meetings are on Microsoft Teams, and documents and notes are shared on SharePoint.

6.2. The Biggest Challenges

The first data challenge is the consistency of the source material. HIQA inspection reports vary in structure, wording, and level of detail. The "About the designated centre" section, which the amenities extraction uses, is written by each nursing home rather than by the inspector, so the same facility can be described in very different ways across the Dublin centres processed. Before extracted amenities can be shown on the site, the model's output is being validated against the curated gold set dataset, which means reading the reports closely and separating what is stated, what is missing, and what cannot be confirmed either way.

A harder problem sits on the compliance side. When a provider commits to fixing a non-compliance, the data does not say whether the commitment was kept. The answer has to be traced through later reports, which often do not re-assess the original finding and sometimes do not mention it at all. In the Dublin test, 36% of findings were not re-assessed in a later report. Building this link between reports, and measuring it against human labels, is the research core of the project and the part at greatest risk of delay.

Evaluation with real users is also difficult. Choosing a nursing home is a sensitive matter, so families and other end users are hard to recruit, which limits how much direct feedback can be gathered.

On the engineering side, integrating the amenities data into the shared database alongside the compliance data is still in progress. Local setup is another source of friction. Getting the project running involves more than cloning the repository, because each developer's database also needs to hold roughly the same data as everyone else's. At the moment, populating a local database depends on running the full backend workflow, which means one developer can end up waiting on

another. We want developers to be able to load the required data independently. Finally, every pull request needs a teammate’s approval before it is merged, so our weekly progress depends on review capacity as much as on the time spent writing code.

6.3. The Remaining Time

Feature development in the remaining period concentrates on four areas. The first is geocoding of the registered address, so that a center can be searched for and shown in relation to the essential services and amenities in its surrounding area rather than by county name alone. The second is an alert system: a user who registers an area is notified when an inspection has taken place and a report published for a center in that area, so that a change in a home's record reaches the family rather than having to be looked for. The third is the comparison grid set out in the earlier wireframes, which places several homes side by side with one regulation per row and each home's judgments in sequence, so that a shortlist can be read briefly. The fourth is natural language searching across the inspection reports, where a question is answered from the report's text, and each answer carries the report and page it came from.

The plan is also to go forward with public announcements brought forward within the same period. The site will be announced once the core features are developed and stable to be used by the end users. The announcement will also provide the first evidence of use by people outside the project team.

7. References and Key Resources

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